Individual Assignment
Weightage: 80%, 100 marks

Introduction to Computer Science - Part 1 **(50 marks)**

Construct a solution algorithm for the following problems. Your solution should contain:

- A defining problem
- A pseudo code algorithm
- A desk check of the algorithm
    - Three test cases for each question. Two valid and one error test case

Question 1

The Express Delivery Service provides the fastest doorstep collection and delivery within 2-hour in Singapore. The delivery charge is calculated based on the zonal classification in Singapore and there are two main types of zones, one is Central Business District (CBD) with the postal codes starting with 01 to 08, 17 to 19, 22 and 23 and other one is NON-CBD for all other areas other than CBD. The postal code in Singapore contains exactly 6 numeric digits. Below mapping table is used to calculate the delivery charge,

Category	Delivery Charge	Weight
Within CBD	$15.00	Up to 5kg
CBD to NON-CBD / NON-CBD to CBD	$20.00	Up to 5kg
Outside CBD	$25.00	Up to 5kg

Design an algorithm that will prompt for and receive the postal code for both pickup and delivery location and calculate the delivery charge and print it. The program will then prompt for "*Do you want to calculate for another delivery (Yes/No)?*" and is to repeat the processing until "No" is entered.

Question 2

An Individual income tax is calculated based on the total taxable income of an individual in a year. The tax rate is defined based on the taxable income table shown below.

	Taxable Income ($)	Rate (%)
On the first	20,000	0
On the next	20,000	2.0
On the next	30,000	3.5
On the next	40,000	7.0
Above	110,000	11.5

For example, if John's taxable income is $65,000 in 2021, his income tax would be $1275 based on the below calculation,

John's Income Tax for 2021 = (20000 x 0%) + (20000 x 2%) + (25000 x 3.5%)
= 0 + 400 + 875
= $1275

Taxpayer's details are maintained in a file that contains National Identity ID, Full Name and Taxable income for the year 2021. Design an algorithm that will read the taxpayer's details and calculate the income tax for each taxpayer in the file and print the full name and income tax.

Question 3

In an organisation, Employee's bonus is computed based on the company performance and employee performance.

Bonus multiplier for employee's performance rating is defined in the table below.

Employee's Performance Rating	Description	Bonus Multiplier
1	Require Improvement	0.8
2	Achieved	1
3	Exceed	1.5
4	Exceptional	2

Bonus multiplier for company performance is defined in the table below,

Company Performance	Bonus Multiplier
<100%	0.8
100-105%	1
>105%	1.5

Employee's bonus is computed using the formula below,

Employee's Bonus = Employee's Salary X Bonus Multiplier for Employee's Performance X Bonus Multiplier for Company Performance

Employees' performance details are maintained in a file that contains employee ID, employee name, employee salary, and employee performance rating (1, 2, 3 or 4). Design an algorithm that will prompt for and receive the company performance in percentage and compute the bonus for each employee in the file and print the employee ID, employee name and bonus.

Introduction to Computer Science - Part 2 **(50 marks)**

Implement the three questions algorithm in Part 1 into Java programming. Your code should contain appropriate validations and must focus on code optimisation.

You need to submit:

- Three Java code
- For each question explain (100-150 words) how the logic works.
- Three sample output screenshots
    - Two normal and one error test cases